

Simplified Denitrification Example with Calcite

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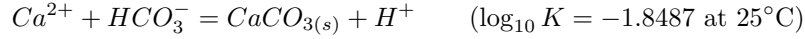
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1 Chemistry

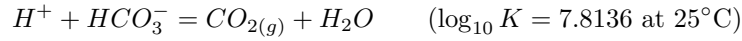
We consider a simplified denitrification process in a clean aquifer, where polluted water enters through the inflow. We only consider two redox reactions: aerobic oxidation of DOC and denitrification from nitrate to nitrogen, thus neglecting the entire nitrate reduction sequence. The aqueous species are Cl^- , $CH_2O_{(aq)}$, H^+ , NO_3^- , $N_{2(aq)}$, Ca^{2+} , HCO_3^- , $O_{2(aq)}$, CO_3^{2-} , and H_2O . The gases are $CO_{2(g)}$ and $O_{2(g)}$, and the mineral phase is calcite, $CaCO_{3(s)}$. We assume that there is bicarbonate dissociation and that the water is in equilibrium with calcite both in the aquifer and at the inflow. We also assume that water entering the aquifer is in equilibrium with atmospheric O_2 and CO_2 .

We consider the following equilibrium reactions.

- Calcite dissolution/precipitation:



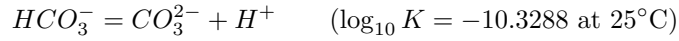
- Atmospheric CO_2 :



- Atmospheric O_2 :

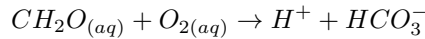


- Bicarbonate dissociation:

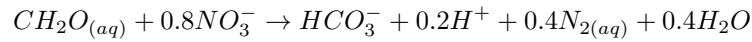


and kinetic reactions:

- Aerobic oxidation of DOC:



- Denitrification ($NO_3^- \rightarrow N_{2(aq)}$):



The kinetic reaction rates are governed by the Monod formula ([Monod \(2012\)](#)):

$$\begin{cases} r_{K,1} = \mu_1 \frac{[CH_2O_{(aq)}]}{k_{DOC} + [CH_2O_{(aq)}]} \frac{[O_{2(aq)}]}{k_{O_2} + [O_{2(aq)}]} \\ r_{K,2} = \mu_2 \frac{I_{O_2}}{I_{O_2} + [O_{2(aq)}]} \frac{[CH_2O_{(aq)}]}{k_{DOC} + [CH_2O_{(aq)}]} \frac{[NO_3^-]}{k_{NO_3^-} + [NO_3^-]} \end{cases}$$

where the brackets indicate concentrations, μ are the maximum reaction rate constants, k are the half-saturation constants, and I are the inhibition constants. The parameter values used in this example are:

Parameter	Reaction 1	Reaction 2
μ (mol/L/d)	$5 \cdot 10^{-1}$	2
k_{DOC} (mol/L)	10^{-3}	$1.4 \cdot 10^{-3}$
k_{O_2} (mol/L)	10^{-5}	—
$k_{NO_3^-}$ (mol/L)	—	$5.9 \cdot 10^{-3}$
I_{O_2} (mol/L)	—	$2.52 \cdot 10^{-5}$

Table 1: Monod kinetic parameters

The boundary water is in equilibrium with atmospheric O_2 , atmospheric CO_2 , and calcite, therefore the stoichiometric matrix at the inflow is

$$\mathbf{S}_e^{inf} = \begin{pmatrix} Cl^- & CH_2O_{(aq)} & H^+ & NO_3^- & N_{2(aq)} & Ca^{2+} & HCO_3^- & O_{2(aq)} & CO_3^{2-} & H_2O & CaCO_{3(s)} & CO_{2(g)} & O_{2(g)} \\ 0 & 0 & 1 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

since there are no kinetic reactions. In the domain, the same bicarbonate dissociation and calcite reactions occur, therefore the equilibrium stoichiometric matrix in the domain is

$$\mathbf{S}_e^{dom} = \begin{pmatrix} Cl^- & CH_2O_{(aq)} & H^+ & NO_3^- & N_{2(aq)} & Ca^{2+} & HCO_3^- & O_{2(aq)} & CO_3^{2-} & H_2O & CaCO_{3(s)} \\ 0 & 0 & 1 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \end{pmatrix}$$

and the kinetic stoichiometric matrix in the domain is

$$\mathbf{S}_K = \begin{pmatrix} Cl^- & CH_2O_{(aq)} & H^+ & NO_3^- & N_{2(aq)} & Ca^{2+} & HCO_3^- & O_{2(aq)} & CO_3^{2-} & H_2O & CaCO_{3(s)} \\ 0 & -1 & 1 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0.2 & -0.8 & 0.4 & 0 & 1 & 0 & 0 & 0.4 & 0 \end{pmatrix}.$$

1.1 Component matrix

Following De Simoni et al. (2005), the component matrix $\mathbf{U}$ is defined such that $\mathbf{U}\mathbf{S}_{e,v}^T = \mathbf{0}$, ensuring that the components $\mathbf{u} = \mathbf{U}\mathbf{c}$ are conservative quantities not affected by equilibrium reactions. Here $\mathbf{S}_{e,v}$ denotes the equilibrium stoichiometric matrix restricted to variable activity species (i.e. excluding constant activity species). In the domain H_2O and $CaCO_{3(s)}$ have constant activity (no gaseous phase is present, and the mineral phase has unit activity), whereas at the inflow H_2O , $CO_{2(g)}$, $O_{2(g)}$ and $CaCO_{3(s)}$ all have constant activity (the gases are fixed by atmospheric partial pressures and the mineral has unit activity). In the domain, $\mathbf{S}_{e,v}$ is

$$\mathbf{S}_{e,v} = \begin{pmatrix} Cl^- & CH_2O_{(aq)} & H^+ & NO_3^- & N_{2(aq)} & Ca^{2+} & HCO_3^- & O_{2(aq)} & CO_3^{2-} \\ 0 & 0 & 1 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & 1 \end{pmatrix},$$

obtained from $\mathbf{S}_e^{dom}$ by removing the columns corresponding to H_2O and $CaCO_{3(s)}$.

Splitting $\mathbf{S}_{e,v} = (\mathbf{S}'_{e,v} | \mathbf{S}''_{e,v})$ into blocks corresponding to primary and secondary variable activity species, and redefining so that $\mathbf{S}''_{e,v} = -\mathbf{I}$, the component matrix takes the form

$$\mathbf{U} = (\mathbf{I}_{N_u} | \mathbf{S}'_{e,v}^T), \quad (1.1)$$

where $N_u = N_v - N_r$ is the number of components, N_v is the number of variable activity species and N_r the number of equilibrium reactions.

In the domain, we have $N_v = 9$ variable activity species (Cl^- , $CH_2O_{(aq)}$, H^+ , NO_3^- , $N_{2(aq)}$, Ca^{2+} , HCO_3^- , $O_{2(aq)}$, CO_3^{2-}) and $N_r = 2$ equilibrium reactions (calcite dissolution and bicarbonate dissociation, in that order), giving $N_u = 9 - 2 = 7$ primary species. We choose the secondary variable activity species to be HCO_3^- and CO_3^{2-} , whose concentrations are determined by the equilibrium relations. The primary species are therefore Cl^- , $CH_2O_{(aq)}$, H^+ , NO_3^- , $N_{2(aq)}$, Ca^{2+} and $O_{2(aq)}$. The secondary block (columns HCO_3^- and CO_3^{2-} of $\mathbf{S}_{e,v}$) is

$$\mathbf{S}_{e,v}''^{dom} = \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix},$$

so we redefine $\mathbf{S}_{e,v}'' = -\mathbf{I}$ by premultiplying by $P = -(\mathbf{S}_{e,v}'^{dom})^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$ – i.e. leaving the calcite row unchanged and replacing the bicarbonate dissociation row by the calcite row minus itself – yielding the transformed primary block (columns Cl^- , $CH_2O_{(aq)}$, H^+ , NO_3^- , $N_{2(aq)}$, Ca^{2+} , $O_{2(aq)}$)

$$\mathbf{S}_{e,v}'^{dom} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}.$$

The component matrix is therefore $\mathbf{U}^{dom} \in \mathbb{R}^{7 \times 9}$:

$$\mathbf{U}^{dom} = \left(\mathbf{I} \mid \mathbf{U}_2^{dom} \right)$$

where, with rows corresponding to the primary species Cl^- , $CH_2O_{(aq)}$, H^+ , NO_3^- , $N_{2(aq)}$, Ca^{2+} , $O_{2(aq)}$,

$$\mathbf{U}_2^{dom} = (\mathbf{S}_{e,v}'^{dom})^T = \begin{pmatrix} HCO_3^- & CO_3^{2-} \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ -1 & -1 \\ 0 & 0 \end{pmatrix}.$$

Written out explicitly in the natural species order,

$$\mathbf{U}^{dom} = \begin{pmatrix} Cl^- & CH_2O_{(aq)} & H^+ & NO_3^- & N_{2(aq)} & Ca^{2+} & HCO_3^- & O_{2(aq)} & CO_3^{2-} \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix},$$

which yields the conservative components $u_1 = c_{Cl^-}$, $u_2 = c_{CH_2O_{(aq)}}$, $u_3 = c_{H^+} + c_{HCO_3^-}$, $u_4 = c_{NO_3^-}$, $u_5 = c_{N_{2(aq)}}$, $u_6 = c_{Ca^{2+}} - c_{HCO_3^-} - c_{CO_3^{2-}}$, and $u_7 = c_{O_{2(aq)}}$. The component u_3 is the total proton-bicarbonate alkalinity, and u_6 is the calcium balance corrected by the dissolved carbonate species, ensuring that any change in $c_{Ca^{2+}}$ due to calcite dissolution/precipitation is compensated by an equal but opposite change in $c_{HCO_3^-} + c_{CO_3^{2-}}$, as required by mass conservation.

Component matrix of the boundary (inflow) water. At the inflow, all four equilibrium reactions apply, so $N_r = 4$. Since H_2O , $CO_{2(g)}$, $O_{2(g)}$, and $CaCO_{3(s)}$ all have constant activity, $\mathbf{S}_{e,v}^{inf}$ is obtained from $\mathbf{S}_e^{inf}$ by removing these four columns, leaving the same nine variable activity species as in the domain (Cl^- , $CH_2O_{(aq)}$, H^+ , NO_3^- , $N_{2(aq)}$, Ca^{2+} , HCO_3^- , $O_{2(aq)}$, CO_3^{2-}), so $N_v = 9$ and $N_u = 9 - 4 = 5$ components. The matrix is

$$\mathbf{S}_{e,v}^{inf} = \begin{pmatrix} Cl^- & CH_2O_{(aq)} & H^+ & NO_3^- & N_{2(aq)} & Ca^{2+} & HCO_3^- & O_{2(aq)} & CO_3^{2-} \\ 0 & 0 & 1 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & 1 \end{pmatrix},$$

where the rows correspond to the calcite, atmospheric CO_2 , atmospheric O_2 , and bicarbonate dissociation reactions, in that order. We choose secondary variable activity species Ca^{2+} , HCO_3^- , $O_{2(aq)}$, and CO_3^{2-} (one per equilibrium reaction, in the same row order as the reactions), so the primary species are Cl^- , $CH_2O_{(aq)}$, H^+ , NO_3^- , and $N_{2(aq)}$. The secondary block (columns Ca^{2+} , HCO_3^- , $O_{2(aq)}$, CO_3^{2-} of $\mathbf{S}_{e,v}^{inf}$) is

$$\mathbf{S}_{e,v}''^{inf} = \begin{pmatrix} -1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix},$$

which is not $-\mathbf{I}$ because the calcite row has non-zero entries for both Ca^{2+} and HCO_3^- , and the bicarbonate dissociation row has non-zero entries for both HCO_3^- and CO_3^{2-} . Redefining $\mathbf{S}_{e,v}'' = -\mathbf{I}$ by premultiplying by $P = -(\mathbf{S}_{e,v}'')^{-1} =$

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} - \text{i.e. replacing the calcite row by the calcite row minus the atmospheric } CO_2 \text{ row, leaving the}$$

atmospheric CO_2 and O_2 rows unchanged, and replacing the bicarbonate dissociation row by the atmospheric CO_2 row minus itself – yields the transformed primary block (columns Cl^- , $CH_2O_{(aq)}$, H^+ , NO_3^- , $N_{2(aq)}$)

$$\mathbf{S}_{e,v}^{inf} = \begin{pmatrix} 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \end{pmatrix}.$$

The component matrix at the inflow is $\mathbf{U}^{inf} \in \mathbb{R}^{5 \times 9}$:

$$\mathbf{U}^{inf} = (\mathbf{I} \mid \mathbf{U}_2^{inf})$$

where, with rows corresponding to the primary species Cl^- , $CH_2O_{(aq)}$, H^+ , NO_3^- , $N_{2(aq)}$,

$$\mathbf{U}_2^{inf} = (\mathbf{S}_{e,v}^{inf})^T = \begin{pmatrix} Ca^{2+} & HCO_3^- & O_{2(aq)} & CO_3^{2-} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & -1 & 0 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Written out explicitly in the natural species order,

$$\mathbf{U}^{inf} = \begin{pmatrix} Cl^- & CH_2O_{(aq)} & H^+ & NO_3^- & N_{2(aq)} & Ca^{2+} & HCO_3^- & O_{2(aq)} & CO_3^{2-} \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 2 & -1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

It is straightforward to verify that $\mathbf{U}^{inf}(\mathbf{S}_{e,v}^{inf})^T = \mathbf{0}$. The five components are $u_1^{inf} = c_{Cl^-}$, $u_2^{inf} = c_{CH_2O_{(aq)}}$, $u_3^{inf} = c_{H^+} - c_{HCO_3^-} - 2c_{CO_3^{2-}} + 2c_{Ca^{2+}}$, $u_4^{inf} = c_{NO_3^-}$, and $u_5^{inf} = c_{N_{2(aq)}}$. Note that $O_{2(aq)}$ does not appear in any component: its concentration is directly fixed by the atmospheric O_2 equilibrium ($c_{O_{2(aq)}} = K_{O_2}^{-1}a_{O_{2(g)}}$), so it carries no degree of freedom. Likewise, the atmospheric CO_2 equilibrium combined with bicarbonate dissociation and calcite equilibrium reduces the four carbonate/calcium-related species H^+ , Ca^{2+} , HCO_3^- , CO_3^{2-} to a single component. The boundary water therefore has only five independent components, in contrast with the seven components of the domain, reflecting the two additional constraints imposed by equilibrium with the atmospheric gases.

Note on the component matrix used in the simulation. Although two component matrices have been derived above ($\mathbf{U}^{dom}$ and $\mathbf{U}^{inf}$), the reactive transport simulation uses the *domain* component matrix $\mathbf{U}^{dom}$, not the boundary one. This is because the components $\mathbf{u} = \mathbf{U}\mathbf{c}$ must be conservative throughout the model domain, where only the calcite and bicarbonate dissociation equilibria are active; the additional atmospheric gas equilibria at the inflow are not enforced inside the aquifer. The boundary component matrix $\mathbf{U}^{inf}$ is used only to speciate the inflow water (i.e. to compute the boundary concentrations consistent with atmospheric and calcite equilibrium), and the resulting concentrations $\mathbf{c}_{inf}$ are then projected onto the domain components via $\mathbf{u}_{inf} = \mathbf{U}^{dom}\mathbf{c}_{inf}$ before being imposed as a boundary condition for the transport of components. More generally, the component concentrations $\mathbf{u} = \mathbf{U}^{dom}\mathbf{c}$ used in the reactive transport simulation are *always* computed with the domain component matrix, regardless of whether the underlying species concentrations $\mathbf{c}$ correspond to the domain (initial or interior) or to the boundary (inflow) water.

The initial and boundary concentrations we consider are in Table 2:

	initial	inflow
Cl^-	10^{-16}	10^{-3}
$CH_2O_{(aq)}$	10^{-5}	$2 \cdot 10^{-3}$
H^+	10^{-7}	10^{-8}
NO_3^-	10^{-5}	10^{-3}
$N_{2(aq)}$	10^{-16}	10^{-16}
Ca^{2+}	$7.0583 \cdot 10^{-3}$	$1.3129 \cdot 10^{-3}$
HCO_3^-	10^{-3}	$5.3761 \cdot 10^{-4}$
$O_{2(aq)}$	10^{-6}	$2.5295 \cdot 10^{-4}$

Table 2: Primary species concentrations of initial and boundary waters (in molalities). The Ca^{2+} concentrations are imposed by equilibrium with calcite at the given H^+ and HCO_3^- activities.

2 Transport

We solve the following set of equations:

$$\begin{cases} \phi \frac{\partial \mathbf{c}^T}{\partial t} = \mathcal{L}^T(\mathbf{c}) + \phi \mathbf{r}^T \\ \mathbf{S}_e \log \mathbf{c} = \log \mathbf{K}_e \\ \mathbf{r}_K = \mathbf{r}_K(\mathbf{c}) \end{cases} \quad (2.1)$$

where $\phi \in (0, 1]$ is porosity; $\mathbf{c} \in \mathbb{R}^{n_s}$ is the concentration array of all species (n_s is the number of species); $\mathcal{L}^T(\mathbf{c}) = -\mathbf{q}^T \nabla \mathbf{c}^T + \nabla \cdot (\mathbf{D} \nabla \mathbf{c}^T) + f_w (\mathbf{c}_r^T - \mathbf{c}^T)$ is the transposed advective form of the transport operator, if we solve transport using the advective ADE, where $\mathbf{q} \in \mathbb{R}^d$ is the Darcy flux (d is the Euclidean dimension, in this case $d = 1$), $\mathbf{D} \in \mathbb{R}^{d \times d}$ is the hydrodynamic dispersion tensor, f_w is the (volumetric) water sink-source, with concentration $\mathbf{c}_r$ if $f_w > 0$ (recharge), or, $\mathbf{c}_r = \mathbf{c}$ if $f_w < 0$ (discharge), or $\mathbf{c}_r = 0$ when $f_w < 0$ represents evaporation; $\mathbf{r} = \mathbf{S}_e^T \mathbf{r}_e + \mathbf{S}_K^T \mathbf{r}_K$ is the contribution of chemical reactions to the mass balance of species (per unit time); $\mathbf{S}_e \in \mathbb{R}^{n_e \times n_s}$ and $\mathbf{S}_K \in \mathbb{R}^{n_K \times n_s}$ are the stoichiometric matrices of equilibrium and kinetic reactions, respectively, whose respective reaction rates are given by $\mathbf{r}_e \in \mathbb{R}^{n_e}$ and $\mathbf{r}_K \in \mathbb{R}^{n_K}$ (n_e and n_K are the number of equilibrium and kinetic reactions, respectively) and $\mathbf{K}_e \in \mathbb{R}^{n_e}$ are the equilibrium constants.

Equation (2.1) needs to be solved subject to appropriate initial ($\mathbf{c}(\mathbf{x}, t) = \mathbf{c}(\mathbf{x}, 0)$ throughout the model domain) and boundary conditions, which we choose to be prescribed mass flux, that is,

$$\begin{cases} \mathbf{n}^T (\mathbf{q} \mathbf{c} - \mathbf{D} \nabla \mathbf{c}) = q_{inf} \mathbf{c}_{inf} & \text{at the inflow} \\ \mathbf{n}^T (-\mathbf{D} \nabla \mathbf{c}) = 0 & \text{at the outflow} \end{cases}$$

expressing that the flux of solute mass at the inflow is $q_{inf} \mathbf{c}_{inf}$, and that there is no dispersive flux at the outflow.

To solve numerically, we choose a uniform mesh made of 10 cells of length $\Delta x = 0.1$ m (domain length $L = 1$ m). The time step is $\Delta t = 10^{-2}$ days. We use the fully implicit Euler scheme for both transport and reactions. We consider uniform transport properties: $\phi = 0.5$, $q = q_{inf} = 0.2m/d$, $\alpha_L = 1m$. The dispersion coefficient is $D = \alpha_L q = 0.2m^2/d$. Note that while the chemistry described in Section 1 (species, reactions, stoichiometry, equilibrium constants, and Monod parameters) is intrinsic to this denitrification problem, the transport parameters and numerical settings — mesh size, time step, porosity, Darcy flux, dispersivity, boundary condition type, and integration scheme — can be freely modified by the user to suit different physical scenarios, subject to the following stability and accuracy constraints:

- **Grid Péclet number:** $Pe = \frac{q \Delta x}{D} = \frac{\Delta x}{\alpha_L} \leq 2$ to avoid spurious oscillations in centred schemes. In this example, $Pe = 0.1$.
- **Courant number:** $Co = \frac{q \Delta t}{\phi \Delta x} \leq 1$ for time integration. In this example, $Co = 0.2 \cdot 10^{-2} / (0.5 \cdot 0.1) = 0.04$.
- **Von Neumann stability:** For the explicit Euler scheme, the time step must satisfy $\Delta t \leq \frac{\phi(\Delta x)^2}{2D}$ for dispersion stability. The fully implicit Euler scheme is unconditionally stable, so this constraint does not apply.
- **Combined constraint (explicit Euler):** When using explicit time integration, both the Courant and dispersion constraints must be satisfied simultaneously: $\Delta t \leq \min \left(\frac{\phi \Delta x}{q}, \frac{\phi(\Delta x)^2}{2D} \right)$.

The fully implicit Euler scheme used in this example is unconditionally stable, allowing larger time steps at the cost of solving a nonlinear system at each step. However, accuracy still requires Δt small enough to resolve the reaction dynamics.

116 **References**

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